

Multi-Agent Flood Early-Warning Portal

Comprehensive Technical & Operational Report

National Disaster Management Authority (NDMA), Pakistan

Report generated	10 July 2026, 09:40 UTC
System status	Operational (prototype) — all agents live
Data cost	PKR 0 — 100% free & open official sources
Classification	Official — internal briefing

1. Executive Summary

NDMA's Flood Early-Warning Portal is a working prototype that fuses six autonomous software agents — weather, river discharge, terrain, population exposure, urban flooding and a coordinating decision engine — into a single operational WebGIS. Every number the system displays is computed live from official, free international data archives (NASA, NOAA, Copernicus, USGS, WorldPop, FAO). Nothing is licensed, nothing is manually entered, and nothing is invented by artificial intelligence: the risk score and alert decision are produced by transparent, auditable rules, while an AI assistant named **Alpha** only orchestrates and explains those results.

The portal answers, in one screen: *how much rain has fallen and is coming; what the rivers will do over the next 30 days of forecast horizon; who is exposed; which cities face street flooding; and what action level is warranted* — for any river basin, any province or district of Pakistan (including Gilgit-Baltistan and Azad Kashmir), or any rectangle drawn by hand on the map.

Key differentiators

- **Zero data cost:** every layer and figure comes from free official archives; the only account required is a Google Earth Engine service account (free for government/research use).
- **Deterministic risk, explainable end-to-end:** rule-based fusion with published weights and thresholds calibrated on Pakistan's 2010 and 2022 floods; the AI never sets the score.
- **Full national coverage:** 5 river basins, 8 provinces/territories, 100+ districts, 17 major cities, and arbitrary user-drawn areas — all through one resolver.
- **Conversational operations:** the Alpha assistant runs any pipeline by voice or text, drives the map, and reports agents' exact numbers, with automatic fail-over across three independent AI providers including a fully offline local model.
- **Honest engineering:** every known limitation (satellite latency, GloFAS bias, proxy indicators) is documented, displayed, and carried into the decision logic rather than hidden.

2. Current Situation Snapshot (live system output)

All figures below were produced by the running system at report time (2026-07-10 08:51:58 UTC) — they are pasted verbatim from agent output, exactly as the portal displays them.

2.1 Chenab basin — full multi-agent risk assessment

Indicator	Live value	Interpretation
Fused risk score	0/100 — LOW	Rule-based fusion of rain (obs+fcst) and river components
Decision	routine monitoring	No alert. Continue scheduled monitoring runs.
Observed rain, 72 h (GPM IMERG)	10.0 mm basin mean · 78.3 mm max	Below all alert thresholds
Forecast rain, 72 h (NOAA GFS)	11.4 mm mean · 50.8 mm max	Monsoon activity present but sub-threshold
Air temperature now (GFS 2 m)	35.7 °C AOI mean · 46.6 °C max	Model area value, not a station reading
Worst river station (GloFAS)	Trimmu — 1,806 m³/s peak forecast	Category: normal
Population in basin (WorldPop)	69,085,855	2020 constrained baseline
Of which on floodplain	51,035,192	Slope-proxy floodplain definition
Basin area (HydroSHEDS L6)	88,788 km²	Level-6 sub-basin union

2.2 Urban flood indicator — nationwide city scan

Cities flagged at report time: **Muzaffarabad**. Observed-rain latency: 29.4 h (satellite publication delay, reported transparently).

City	Province / territory	Obs 24 h (mm)	Fcst 24 h (mm)	Indicator
Muzaffarabad	Azad Kashmir	20.4	2.0	WATCH
Gujranwala	Punjab	5.4	0.4	none
Gilgit	Gilgit-Baltistan	0.2	1.8	none
Sialkot	Punjab	1.5	0.2	none
Faisalabad	Punjab	1.1	0.1	none
Lahore	Punjab	0	0.7	none
Karachi	Sindh	0	0.6	none
Bahawalpur	Punjab	0.6	0	none
Quetta	Balochistan	0.2	0	none
Rawalpindi-Islamabad	Punjab	0.1	0	none
Hyderabad	Sindh	0	0	none
Sukkur	Sindh	0	0	none
Multan	Punjab	0	0	none
Dera Ghazi Khan	Punjab	0	0	none
Peshawar	Khyber Pakhtunkhwa	0	0	none
Nowshera	Khyber Pakhtunkhwa	0	0	none
Dera Ismail Khan	Khyber Pakhtunkhwa	0	0	none

Thresholds (mm / 24 h over a 12 km city footprint): <20 none · 20–50 watch (ponding possible) · 50–80 likely (street flooding expected) · ≥80 severe. Aligned with PMD heavy / very-heavy rainfall categories; Karachi drainage studies show street flooding from ~40–50 mm/24 h.

3. What the Portal Is

The portal is a full-screen WebGIS (dark, control-room styled) served by a Python/Flask backend. The analytical brain is a set of independent agents — each one an expert in a single domain, each pulling its own official data source — coordinated by a rule-based fusion engine. The operator interacts three ways:

- **Point-and-click:** pick any area of interest and press one of six pipeline buttons (Weather, Disaster, Terrain, Exposure, Urban flooding, Full risk assessment).
- **Draw on the map:** drag a rectangle anywhere — it instantly becomes a first-class area of interest and every pipeline runs on exactly that box.
- **Ask Alpha:** type or speak a request ("full risk for Sindh", "temperature in Karachi now", "where is urban flooding?") — the assistant runs the real pipelines, moves the map, and answers with the agents' exact figures, optionally aloud in a female voice.

Areas of interest

AOI type	Examples	How resolved
River basins (5)	Chenab, Jhelum, Ravi, Sutlej, Indus	HydroSHEDS level-6 sub-basin unions with registered FFD stations
Provinces / territories (8)	Punjab, Sindh, Balochistan, Khyber Pakhtunkhwa, Gilgit-Baltistan, Azad Kashmir, Islamabad, FATA*	FAO GAUL 2015 (modern names displayed); GB & AJK geometry from geoBoundaries
Districts (100+)	Multan, Tharparkar, Attock ...	FAO GAUL 2015 level-2
Drawn rectangle	any box drawn with the rectangle tool	Converted to a bbox spec; same resolver
Nationwide	urban city scan	17-city registry

*FATA merged into Khyber Pakhtunkhwa in 2018; it remains individually selectable because the boundary dataset carries its geometry separately — noted on-screen rather than hidden.

4. The Six Agents

Agent	Question it answers	Primary data	Update cycle
Weather	How much rain fell (24/72 h), how much is coming (72 h), temperature now	NASA GPM IMERG V07 (observed, ~11 km); NOAA GFS 0.25° (forecast)	IMERG: half-hourly, 1–3 day publication latency; GFS: 4 runs/day
River / Disaster	River discharge forecast at FFD stations, classified into official flood categories	Copernicus GloFAS 30-day ensemble forecast	Daily
Terrain	Elevation, slope, flow accumulation, basin geometry	USGS SRTM 90 m DEM; WFP/HydroSHEDS	Static (cached GeoTIFFs)
Population / Exposure	How many people in the AOI; how many on the floodplain	WorldPop 2020 constrained 100 m	Static baseline
Urban flood	Which major cities face street flooding (17-city scan)	Same IMERG + GFS over 12 km city footprints	With each rain update
Coordinator + Report	Fused 0–100 score, decision ladder, situation report (MD + JSON)	All of the above via message-passing	On demand, seconds

Agents communicate through a logged message-passing pipeline: terrain resolves the AOI, weather's rain outlook is handed to the river agent as context, population quantifies exposure, and the coordinator fuses everything. The full inter-agent conversation is displayed in the portal for every run — a reviewer can read exactly why a score was produced.

5. Risk Methodology — Rules Decide, AI Speaks

5.1 Fusion

The risk score is a weighted combination of three components: **river discharge (50%)**, **forecast rain (30%)** and **observed rain (20%)**. Rain is scored on the 72-hour basin-mean (steps at 25/35/50/70/90 mm); river on each station's official FFD cusec bands (normal → exceptional). If an agent fails or an AOI has no river station, the fusion renormalises around the missing component and marks the assessment *degraded* — it never silently fills gaps.

5.2 Decision ladder

Score	Decision	Meaning
0–24	routine_monitoring	No alert; scheduled runs continue
25–49	advisory	Heightened watch; brief relevant PDMAs
50–74	alert	Pre-position response; issue public advisory
75–100	warning	Activate emergency protocols

5.3 Calibration and backtesting

- Rain steps calibrated on IMERG reconstructions: 2010 flood peak $\approx$ 100 mm/72 h basin-mean, 2022 $\approx$ 51 mm, against a 2018 non-flood control $\approx$ 34 mm.
- Discharge backtest ranks the three years correctly (2010 > 2022 $\approx$ 2018) at Chenab stations.
- Basin-mean was chosen over basin-max deliberately: a single foothill pixel exceeds 200 mm in any monsoon, so max-based scoring would cry wolf continuously.

5.4 The AI boundary

Large language models appear in exactly two places: drafting the human-readable warning text after the rule-based decision is made, and powering the Alpha assistant. In both, hard system rules forbid inventing, estimating or re-computing any figure; every number must be quoted verbatim from an agent result. Chat is architecturally read-only — its tools can run pipelines and read reports, but no tool exists that writes or edits data.

6. Validation Against Independent Sources

6.1 Commercial cross-check (Meteoblue)

On 8 July 2026 the team compared the portal's rainfall fields against an independent commercial provider (Meteoblue precipitation isobands). Within Pakistan all three products agreed (light rain, ≤ 12 mm); apparent disagreements at high bands were fully explained — those bands lay outside Pakistan (76–79°E, in India), and Meteoblue's 4 km model paints sharper convective peaks than the 25 km GFS. Conclusion: no pipeline defect; the portal's fields are consistent with a paid product at zero cost.

6.2 Known bias, stated openly (GloFAS)

Backtesting found GloFAS discharge runs roughly 2× low versus FFD-reported cusecs at Chenab stations (e.g. 2010 Marala: GloFAS $\approx$ 114k vs $\sim$ 260k reported). Relative ranking is correct, so categories ordered by return period remain usable, but absolute thresholds would under-warn. This is documented in the interface and in this report; the roadmap includes per-station bias correction before operational use. We consider stating this openly a strength of the engineering culture behind the portal, not a weakness.

6.3 Continuous transparency mechanisms

- Observed-rain latency (hours) is printed on every weather output — the operator always knows how old the satellite data is.
- Every assessment can be saved as a timestamped situation report (Markdown + JSON) forming an auditable history.
- Each agent's raw JSON is one click away in the interface — nothing is summarised beyond inspection.

7. Portal Features (WebGIS)

Feature	Description
GIS layer catalogue	DEM, slope, river network, observed/forecast rain rasters, rain-by-district and rain-by-province choropleths, population density, admin boundaries — per-layer opacity, search, legends
Six pipeline buttons	Weather · Disaster · Terrain · Exposure · Urban flooding (works with no AOI selected — nationwide) · Full risk assessment
Drawn-box AOI	Rectangle tool: drag a box; it becomes the AOI for every pipeline (capped $\sim 8^\circ \times 8^\circ$)
Searchable AOI picker	Type-ahead search across basins, provinces and 100+ districts; Enter selects the first match
Risk banner & decision	Colour-coded score with the decision ladder output and drafted warning text
Agent conversation	The logged inter-agent hand-offs for the current run, shown as a chat transcript
Report history	One-click save; browsable history of all situation reports (MD + JSON)
Map tools	Zoom, locate, fullscreen, distance measure, dark/satellite basemap toggle
Station markers	FFD stations coloured by live flood category with peak-discharge popups
City markers	17 urban-scan cities coloured none/watch/likely/severe with rain popups

8. Alpha — the AI Operations Assistant

Alpha is a floating, hideable chat panel wired into the same pipeline dispatch as the buttons — one orchestration codebase, no duplication. It speaks with a female voice (browser speech synthesis) and accepts voice input with live transcription.

8.1 What Alpha can do

- Run any pipeline on any AOI by name ("full risk for Sindh", "rain in Tharparkar").
- Drive the live map: zoom, fly to an AOI, switch layers on/off, toggle basemap.
- Answer from the platform's actual datasets, quoting exact agent values with units and latency.
- Retrieve and summarise saved situation reports.

8.2 What Alpha cannot do — by design

- Invent, estimate or recompute any number; alter the risk score or decision.
- Write, edit or delete any dataset, cache or report — no such tool is exposed to it.

8.3 Resilience — three independent AI providers

Priority	Provider	Model	Notes
1	Groq cloud (free tier)	Llama 3.3 70B → 3.1 8B fallback	Fast; automatic model fallback on rate limits, auto-wait on burst limits
2	Local Ollama (this machine)	Llama 3.1 8B	Fully offline, unlimited, no data leaves the premises; slower on CPU
3	Anthropic API	Claude (Opus)	Optional, if a funded key is configured

If one provider fails mid-conversation, the next is tried automatically. The offline tier means the assistant keeps working with no internet-dependent AI at all — relevant for continuity of operations.

9. Complete Data Source Register

Every dataset, its role, licence and cost:

Dataset	Provider	Used for	Cost
GPM IMERG V07 (~11 km, 30-min)	NASA	Observed rainfall (24 h / 72 h windows)	Free
GFS 0.25° forecast	NOAA	Forecast rain (72 h) + 2 m air temperature	Free
GloFAS 30-day river forecast	Copernicus EMS	River discharge at FFD stations	Free (reg.)
SRTM 90 m DEM	USGS/NASA	Elevation, slope	Free
HydroSHEDS basins & rivers	WWF/USGS	Basin geometry, flow accumulation, river network	Free
WorldPop 2020 constrained 100 m	Univ. of Southampton	Population exposure	Free
GAUL 2015 admin boundaries	FAO	Provinces & districts	Free
geoBoundaries CGAZ	William & Mary geoLab	Gilgit-Baltistan & Azad Kashmir geometry	Free
FFD station flood limits	PMD/FFD	Discharge category bands (cusecs)	Public
CARTO / Esri basemaps	CARTO / Esri	Map background tiles	Free tier

Compute: Google Earth Engine (service account, free for research/government use) performs all raster aggregation server-side — the portal machine only needs a normal internet connection, no GIS workstation.

10. Security & Data Integrity

- Chat/AI layer is read-only with respect to all data — enforced by tool design, not by prompt alone.
- Report download endpoints are pattern-guarded against path traversal.
- All credentials live in a git-ignored .env file; the repository template contains placeholders only.
- Risk scores and decisions are reproducible: same inputs → same output, no stochastic step.
- Every saved report is dual-format (human Markdown + machine JSON) with UTC timestamps for audit.
- The local AI tier (Ollama) keeps conversations entirely on-premises when selected.

11. Known Limitations (stated plainly)

Limitation	Impact	Mitigation / roadmap
IMERG observed rain lags 1–3 days	“Observed” is as current as the satellite allows	Latency printed on every output; forecast fills the gap
GloFAS ~2× low vs FFD cusecs	Absolute discharge thresholds under-warn	Relative ranking correct; per-station bias correction planned
Flash floods & coastal flooding not forecast	GloFAS covers major rivers only	Urban rain indicator partially covers cloudburst risk; Sentinel-1 mapping planned
Floodplain population uses a slope proxy	Exposure split is approximate	Replace with Sentinel-1 historical inundation masks
Rain thresholds calibrated on Chenab only	Other basins less tuned	Extend backtests basin-by-basin
Urban indicator is rain-based, not a drainage model	Says “likely”, not “which street”	Honest proxy; no free real-time urban inundation sensor exists

Limitation	Impact	Mitigation / roadmap
WorldPop baseline is 2020	Population slightly understated	Swap in newer releases when published
Non-Chenab FFD cusec bands unverified	Categories approximate outside Chenab	Verify against official FFD table

12. Recommended Next Steps

- **Near term:** verify all FFD station bands against the official table; apply GloFAS per-station bias correction; git-version the codebase with CI tests (7 automated tests already pass).
- **Medium term:** Sentinel-1 SAR inundation mapping for actual flood extent; SMS/email alert dispatch on decision changes; scheduled automatic national runs with report archiving.
- **Longer term:** district-level calibration; integration with PDMA reporting flows; historical event replay mode for training and drills.

13. Closing Statement

The portal demonstrates that a credible, transparent, multi-hazard flood early-warning capability can be assembled for Pakistan at **zero data cost**, with modern AI used responsibly — as an interface, never as the authority. Every figure in this report was generated by the live system and can be reproduced on demand in front of any audience.

Report auto-generated by the portal's reporting tooling on 10 July 2026, 09:40 UTC. Live values in Section 2 reflect system state at generation time.